Optimal Margin Distribution Machine with Sparsity Inducing Penalty

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***Abstract*—Recently a promising research direction of statistical learning has been advocated, i.e., the optimal margin distribution learning, with the central idea of optimizing the margin distribution. As the most representative approach of this new learning paradigm, the optimal margin distribution machine (ODM) considers maximizing the margin mean and minimizing the margin variance simultaneously. The standard ODM exploits the ℓ_2-norm penalty, which gives rise to a dense decision boundary. However, in some situations, the model with parsimonious representation is more preferred, due to the redundant noisy features or limited computing resources. In this paper, we propose the sparse optimal margin distribution machine (Sparse ODM), which aims to achieve better generalization performance with moderate model size. For optimization, we extends an efficient coordinate descent method to solve the final problem since the variables are decoupled. In each iteration, we propose a modified Newton method to solve the one-variable sub-problem. Experimental results on both synthetic and real data sets show the superiority of the proposed method.**

***Index Terms*—margin distribution, sparse, classification**

I. Introduction

Support vector machines (SVMs) and Boosting have always been two mainstream learning approaches during the past decades. The former [1] roots in the statistical learning theory [2] which aims to search a large margin separator in a predefined reproducing kernel Hilbert space, while the latter also enjoys a long history of being explained by margin theory [3], [4] because it is resistant to over-fitting empirically [5]–[7].

Recently, the margin theory for Boosting has finally been defended [8], and discloses that, instead of optimizing a single margin, margin distribution is more crucial to the generalization performance. Later, similar results have also been obtained for SVM-style learning approaches. Specifically, Zhang and Zhou proposed the optimal margin distribution machines (ODMs) for both binary and multi-class classification problems respectively [9]–[11], and achieved superior generalization performance to the traditional large margin based methods. After that, the optimal margin distribution learning has become a promising research direction and attracted a lot of attentions, just to name a few, Zhou and Zhou exploit it to handle unequal misclassification cost [12]; Ou et al. apply it to feature elimination [13]; Zhang and Zhou extends it to clustering and semi-supervised learning ([14], [15]).

The standard ODM exploits the ℓ_2-norm penalty, which gives rise to a dense decision boundary, i.e., a weight vector with no zero entries. However, when redundant noisy features exist, the model with parsimonious representation is more preferred. Even if the underlying problem is not sparse, one often looks for the best sparse approximation to make the model more interpretable, or computationally cheaper to use. In these situations, a sparsity-inducing penalty is often introduced, together with the surrogate loss, making up the objective function. In this paper, we propose the sparse optimal margin distribution machine (Sparse ODM), which aims to find a classification hyperplane optimizing the margin distribution with sparse representation. For optimization, we extends an efficient coordinate descent method to solve the final problem since the variables are decoupled. In each iteration, we propose a modified Newton method to solve the one-variable sub-problem. Extensive experiments on both synthetic and real data sets show that our method is significantly better than the compared methods.

The rest of the paper is organized as follows. We first briefly introduce some preliminaries, and then present our proposed method. After that we reports the empirical studies. Finally we conclude the paper with future works.

II. Preliminaries

To explain the basic idea of ODM, we start with the simple binary classification problem. Let $\mathcal{X} \subseteq \mathbb{R}^n$ and $\mathcal{Y} = \{1, -1\}$ denote the input and output spaces. $\mathcal{D}$ is an unknown distribution over $\mathcal{X} \times \mathcal{Y}$. A training set $\mathcal{S} = \{(\boldsymbol{x}_1, y_1), \ldots, (\boldsymbol{x}_m, y_m)\} \in (\mathcal{X} \times \mathcal{Y})^m$ is drawn identically and independently (i.i.d.) according to $\mathcal{D}$. The hypothesis is defined as the linear function $h(\boldsymbol{x}) = \boldsymbol{w}^\top \boldsymbol{x}$ and the predicted label for unseen instance is the sign of the predicted value, which naturally leads to the definition of margin $\gamma(\boldsymbol{x}, y) = y\boldsymbol{w}^\top \boldsymbol{x}$ [16]. Thus the larger the margin, the more confidence we have on the prediction of $\boldsymbol{x}$'s label, and h misclassifies an instance if and only if it produces a negative margin.

It's well known that SVMs employ the large margin principle to select the decision boundary. As a result, the final classification hyperplane only consists of a small amount of instances [17], i.e., the support vectors (SVs), and the rest

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instances (non-SVs) are totally ignored, which may be misleading in some situations. See Figure 1 for an illustration [18].

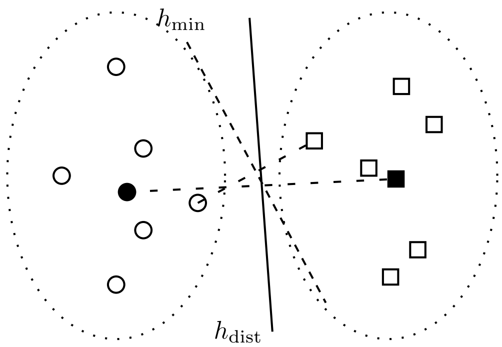

Fig. 1. A simple illustration of linear classifiers optimizing the minimum margin and margin distribution, respectively. Dotted ellipses are two underlying distributions, from which circles and squares are instances sampled. Solid circle and square are mean instances (not necessarily real instances in the training instances). $h_{\min}$ and h_{dist} are classification hyperplanes obtained by optimizing the minimum margin and margin distribution, respectively.

Rather than optimizing a single margin, one more robust strategy is to optimize the margin distribution by exploiting the whole data, which prevents from being cheated by the noisy instances. To characterize the margin distribution, the two most straightforward statistics are the first- and second-order statistics, i.e., the margin mean and variance. Moreover, as suggested in [8], maximizing the margin mean and minimizing the margin variance simultaneously can yield a tighter generalization bound, one can arrive at the following formulation,

$$\begin{aligned} \min_{\boldsymbol{w},\bar{\gamma},\xi_i,\epsilon_i} \ & \Omega(\boldsymbol{w}) - \eta\bar{\gamma} + \frac{\lambda}{m}\sum_{i\in[m]}(\xi_i^2+\epsilon_i^2), \\ \text{s.t. } & y_i\boldsymbol{w}^\top\boldsymbol{x}_i \geq \bar{\gamma} - \xi_i, \\ & y_i\boldsymbol{w}^\top\boldsymbol{x}_i \leq \bar{\gamma} + \epsilon_i, \ \forall i \in [m], \end{aligned} \tag{1}$$

where $\Omega(\boldsymbol{w})$ is the regularization term penalizing the model complexity, $\bar{\gamma}$ is the margin mean, $[m] = \{1, 2, \ldots, m\}$, η and λ are trading-off parameters, ξ_i and ϵ_i are deviations from $\bar{\gamma}$, so it's not difficult to find that the last term $\sum_{i\in[m]}(\xi_i^2+\epsilon_i^2)/m$ is exactly the margin variance.

The Eq. (1) can be simplified by fixing the margin mean as 1 since scaling $\boldsymbol{w}$ doesn't affect the classification results. Besides, introducing different weights for the two deviations can make the method more flexible and characterize the margin distributions more adaptively. Finally, to control the number of SVs, one can tolerate the deviation smaller than a given threshold θ. The formulation of ODM turns to:

$$\begin{aligned} \min_{\boldsymbol{w},\xi_i,\epsilon_i} \ & \Omega(\boldsymbol{w}) + \frac{\lambda}{m}\sum_{i\in[m]}\frac{\xi_i^2+\mu\epsilon_i^2}{(1-\theta)^2}, \\ \text{s.t. } & y_i\boldsymbol{w}^\top\boldsymbol{x}_i \geq 1-\theta-\xi_i, \\ & y_i\boldsymbol{w}^\top\boldsymbol{x}_i \leq 1+\theta+\epsilon_i, \ \forall i \in [m]. \end{aligned} \tag{2}$$

where μ is a parameter for trading-off the two deviations, θ is the threshold for controlling the number of SVs, and $(1-\theta)^2$ in the denominator is to scale the second term to be a surrogate loss for 0-1 loss.

III. SPARSE ODM

The standard ODM applies the ℓ_2-norm regularization term by setting $\Omega(\boldsymbol{w}) = \|\boldsymbol{w}\|_2^2$ in Eq. (2), which is not suitable in some situations, especially when redundant noisy features exist. To achieve a sparse model, a penalty of the cardinality of the support [1] of $\boldsymbol{w}$ is usually adopted. However, this will lead to a difficult mixed-integer programming. A commonly used convex relaxation is to replace the cardinality of the support with the ℓ_1-norm penalty, so we arrive at the Sparse ODM:

$$\begin{aligned} \min_{\boldsymbol{w},\xi_i,\epsilon_i} \ & f(\boldsymbol{w}) = \|\boldsymbol{w}\|_1 + \frac{\lambda}{m}\sum_{i\in[m]}\frac{\xi_i^2+\mu\epsilon_i^2}{(1-\theta)^2}, \\ \text{s.t. } & y_i\boldsymbol{w}^\top\boldsymbol{x}_i \geq 1-\theta-\xi_i, \\ & y_i\boldsymbol{w}^\top\boldsymbol{x}_i \leq 1+\theta+\epsilon_i, \ \forall i \in [m]. \end{aligned} \tag{3}$$

where λ, μ, θ are parameters described in the previous section.

By introducing functions $p_i(\boldsymbol{w}) = 1-\theta-y_i\boldsymbol{w}^\top\boldsymbol{x}_i$ and $q_i(\boldsymbol{w}) = y_i\boldsymbol{w}^\top\boldsymbol{x}_i - 1 - \theta$, the objective function of ODM can be equivalently rewritten as the following unconstrained form:

$$f(\boldsymbol{w}) = \|\boldsymbol{w}\|_1 + \frac{\lambda}{m}\sum_{i\in I_p(\boldsymbol{w})}\frac{p_i^2(\boldsymbol{w})}{(1-\theta)^2} + \frac{\lambda\mu}{m}\sum_{i\in I_q(\boldsymbol{w})}\frac{q_i^2(\boldsymbol{w})}{(1-\theta)^2},$$

where $I_p(\boldsymbol{w}) = \{i \mid p_i(\boldsymbol{w}) > 0\}$ and $I_q(\boldsymbol{w}) = \{i \mid q_i(\boldsymbol{w}) > 0\}$ are index sets indicating a loss occurs.

A. Coordinate Descent Method

Note that the variables are decoupled, we apply the coordinate descent method to solve Sparse ODM. Specifically, the algorithm starts from an initial point $\boldsymbol{w}_1$, and produces a sequence $\{\boldsymbol{w}_1, \boldsymbol{w}_2, \ldots\}$. At the t-th epoch, $\boldsymbol{w}_{t+1}$ is constructed by sequentially updating each component of $\boldsymbol{w}_t$, i.e., each epoch is composed of n iterations. At the j-th iteration, $w_t^{(j)}$ is selected as the variable to minimize while the other variables are kept as constants. This process generates the vectors $\{\boldsymbol{w}_{t,1}, \ldots, \boldsymbol{w}_{t,n}\}$ such that $\boldsymbol{w}_{t,1} = \boldsymbol{w}_t$, $\boldsymbol{w}_{t,n+1} = \boldsymbol{w}_{t+1}$, and

$$\boldsymbol{w}_{t,j} = [w_{t+1}^{(1)}; \ldots; w_{t+1}^{(j-1)}; w_t^{(j)}; w_t^{(j+1)}; \ldots; w_t^{(n)}].$$

For updating $\boldsymbol{w}_{t,j}$ to $\boldsymbol{w}_{t,j+1}$, one needs to solve the following one-variable sub-problem:

$$\min_d \ f(\boldsymbol{w}_{t,j} + d\boldsymbol{e}_j) \tag{4}$$

where $\boldsymbol{e}_j$ is the vector with 1 in the j-th coordinate and 0's elsewhere. To simplify the notations, we denote $\boldsymbol{w}_{t,j}$ as $\boldsymbol{w}$ and define

$$\begin{aligned} l(d;\boldsymbol{w}) = & \frac{\lambda}{m}\sum_{i\in I_p(\boldsymbol{w}+d\boldsymbol{e}_j)}\frac{p_i^2(\boldsymbol{w}+d\boldsymbol{e}_j)}{(1-\theta)^2} \\ & + \frac{\lambda\mu}{m}\sum_{i\in I_q(\boldsymbol{w}+d\boldsymbol{e}_j)}\frac{q_i^2(\boldsymbol{w}+d\boldsymbol{e}_j)}{(1-\theta)^2} \end{aligned}$$

[1] We call the set of non-zeros entries of a vector the support.

$$= \frac{\lambda}{m} \sum_{i \in I_p(\boldsymbol{w}+d\boldsymbol{e}_j)} \frac{(1-\theta-y_i\boldsymbol{w}^\top \boldsymbol{x}_i - y_i d\boldsymbol{e}_j^\top \boldsymbol{x}_i)^2}{(1-\theta)^2}$$
$$+ \frac{\lambda\mu}{m} \sum_{i \in I_q(\boldsymbol{w}+d\boldsymbol{e}_j)} \frac{(y_i\boldsymbol{w}^\top \boldsymbol{x}_i + y_i d\boldsymbol{e}_j^\top \boldsymbol{x}_i - 1 - \theta)^2}{(1-\theta)^2},$$

then the sub-problem (4) can be rewritten as [2]

$$\min_d \; g(d) = |w+d| + l(d;\boldsymbol{w}), \tag{5}$$

where w is the j-th entry of $\boldsymbol{w}$.

B. Modified Newton Method

In any interval of d where the sets $I_p(\boldsymbol{w}+d\boldsymbol{e}_j)$ and $I_q(\boldsymbol{w}+d\boldsymbol{e}_j)$ does not change and the sign of $w+d$ does not flip, the objective function $g(d)$ is quadratic. Therefore, $g(d)$ is a piecewise quadratic function with regard to $d \in \mathbb{R}$. Since Newton method is suitable for quadratic optimization, here we apply it for minimizing $g(d)$.

It's not difficult to find that

$$l'(0;\boldsymbol{w}) = \frac{\lambda}{m} \sum_{i \in I_p(\boldsymbol{w})} \frac{2(1-\theta-y_i\boldsymbol{w}^\top\boldsymbol{x}_i)(-y_i x_{ij})}{(1-\theta)^2}$$
$$+ \frac{\lambda\mu}{m} \sum_{i \in I_q(\boldsymbol{w})} \frac{2(y_i\boldsymbol{w}^\top\boldsymbol{x}_i - 1 - \theta)(y_i x_{ij})}{(1-\theta)^2}$$

is a continuous function, where $x_{ij} = \boldsymbol{e}_j^\top \boldsymbol{x}_i$ is the j-th entry of $\boldsymbol{x}_i$, and

$$l''(0;\boldsymbol{w}) = \frac{\lambda}{m} \sum_{i \in I_p(\boldsymbol{w})} \frac{2x_{ij}^2}{(1-\theta)^2} + \frac{\lambda\mu}{m} \sum_{i \in I_q(\boldsymbol{w})} \frac{2x_{ij}^2}{(1-\theta)^2} \tag{6}$$

is a discontinuous step function. So $l(d;\boldsymbol{w})$ is differentiable but not twice differentiable, and Eq. (6) is a generalized second derivative [19] at $d=0$. Now let's consider the following two cases in turn:

- $w+d \geq 0$, then $g'(d) = l'(d;\boldsymbol{w}) + 1$ and $g''(d) = l''(d;\boldsymbol{w})$. Thus the Newton direction will be $-\frac{l'(d;\boldsymbol{w})+1}{l''(d;\boldsymbol{w})}$ if
$$w - \frac{l'(d;\boldsymbol{w})+1}{l''(d;\boldsymbol{w})} \geq 0 \Longleftrightarrow l'(d;\boldsymbol{w}) + 1 \leq l''(d;\boldsymbol{w})w$$
and otherwise $-w$.
- $w+d \leq 0$, then $g'(d) = l'(d;\boldsymbol{w}) - 1$ and $g''(d) = l''(d;\boldsymbol{w})$. Thus the Newton direction will be $-\frac{l'(d;\boldsymbol{w})-1}{l''(d;\boldsymbol{w})}$ if
$$w - \frac{l'(d;\boldsymbol{w})-1}{l''(d;\boldsymbol{w})} \leq 0 \Longleftrightarrow l'(d;\boldsymbol{w}) - 1 \geq l''(d;\boldsymbol{w})w$$
and otherwise $-w$.

Putting the above together, the Newton direction for minimizing $g(d)$ is

$$s = \begin{cases} -\frac{l'(d;\boldsymbol{w})+1}{l''(d;\boldsymbol{w})} & l'(d;\boldsymbol{w}) + 1 \leq l''(d;\boldsymbol{w})w \\ -\frac{l'(d;\boldsymbol{w})-1}{l''(d;\boldsymbol{w})} & l'(d;\boldsymbol{w}) - 1 \geq l''(d;\boldsymbol{w})w \\ -w & \text{o.w.} \end{cases} \tag{7}$$

[2]Here we omit some constants which have no influence on the optimization.

So a simple Newton method to solve the sub-problem (5) begins with $d_1 = 0$ and iteratively performs update $d_{t+1} = d_t - s$ until $0 \in [l'(d;\boldsymbol{w}) - 1, l'(d;\boldsymbol{w}) + 1]$.

It is proved that under an assumption, the above procedure can terminate in finite steps and solve the sub-problem [20]. Coordinate descent method is known to converge if at each iteration the minimum of the sub-problem can be uniquely attained [21]. Unfortunately, the assumption in [20] may not hold in real cases, so taking the full Newton step may not decrease the function $g(d)$, and a line search procedure should be conducted. Specifically, we check if $s, \beta s, \beta^2 s, \dots$ satisfy the following sufficient decrease condition in turn:

$$\begin{aligned} &|w+\beta^t s| - |w| + l(\beta^t s;\boldsymbol{w}) - l(0;\boldsymbol{w}) \\ &\qquad \leq \sigma\beta^t (l'(0;\boldsymbol{w})s + |w+s| - |w|) \end{aligned} \tag{8}$$

where $t = 0,1,2,\dots$, $\beta \in (0,1)$, $\sigma \in (0,1)$, and uses the satisfied one as Newton step.

Algorithm 1 summarizes the pseudo-code of the coordinate descent method for solving Sparse ODM.

Algorithm 1 Coordinate descent method for Sparse ODM

Require: $\boldsymbol{w}_1 \in \mathbb{R}^n$, $\beta = 0.5$, $\sigma = 0.01$.

while $\boldsymbol{w}$ is not optimal **do**
  for $j = 1, \dots, n$ **do**
    Find the Newton direction s according to (7).
    while $t = 0, 1, 2, \dots$ **do**
      if Eq. (8) is satisfied **then**
        Update $w_j \leftarrow w_j + \beta^t s$.
        Break.
      end if
    end while
  end for
end while

IV. Empirical Studies

In this section, we empirically validate the performance of our proposed method on both synthetic and real data sets.

A. Results on Synthetic Data Sets

We generate 50 training instances for each of the two classes. The first class has two standard normal independent features x_1, x_2 . The second class also has two standard normal independent features, but conditioned on $4.5 \leq x_1^2 + x_2^2 \leq 8$. To make the problem harder, we sequentially augment the features with additional 2, 4, 6 and 8 standard normal noise features. Hence the second class almost completely surrounds the first. The Bayes error rate for this problem is 0.0435, irrespective of feature dimension. In the original input space, a hyperplane can not separate the classes, so we use an enlarged feature space corresponding the 2nd degree polynomial kernel, hence the dictionary of basis functions is $D = \{x_i^2, \sqrt{2}x_i x_j, \sqrt{2}x_i, i, j \in [n]\}$. We generate 1000 test instances to compare the ℓ_1-norm SVM and the standard ℓ_2-norm ODM. The average test errors over 10 simulations, with different numbers of noise features, are shown in Table I.

TABLE I
CLASSIFICATION ERROR RATES OF SPARSE ODM AND OTHER COMPARED METHODS ON SYNTHETIC DATA SETS

Synthetic data sets	$\|D\|$	No penalty SVM	ℓ_1-norm SVM	ℓ_2-norm ODM	Sparse ODM
No noise features	5	0.08±0.01	0.077±0.010	**0.071±0.02**	0.073±0.04
2 noise features	14	0.12±0.03	0.09±0.02	0.10±0.04	**0.08±0.02**
4 noise features	27	0.20±0.05	0.12±0.03	0.15±0.01	**0.11±0.01**
6 noise features	44	0.22±0.06	0.13±0.02	0.18±0.01	**0.12±0.02**
8 noise features	65	0.22±0.06	0.14±0.05	0.19±0.01	**0.11±0.03**

For both the Sparse ODM and other baselines, we tune the parameters according to 5-fold cross validation, to be as fair as possible to each method. For comparison, we also include the results of the standard non-penalized SVM.

From Table I we can see that the non-penalized SVM performs significantly worse than the penalized ones. The ℓ_2-norm ODM obtain the best accuracy when there is no noise features, but is adversely affected by noise features. Since the ℓ_1-norm SVM and Sparse ODM have the ability to select relevant features and ignore redundant features, they do not suffer from the noise features as much as the ℓ_2-norm ODM.

B. Results on Real Data Sets

We apply the Sparse ODM to leukemia data set [22]. This data set consists of 38 training instances and 34 test instances. Each instance is a vector with $n = 7,129$ features. We use the original features as the basis functions. The parameters is tuned according to 5-fold cross validation, then the classifier is trained on all the training instances and evaluated on the test instances. The results are summarized in Table II. We can see that the Sparse ODM performs better than other methods, and it has the advantage of automatically selecting relevant features.

TABLE II
CLASSIFICATION RESULTS OF SPARSE ODM AND OTHER COMPARED METHODS ON LEUKEMIA DATA SET.

Methods	CV Error	Test Error	# selected features
ℓ_1-norm SVM	3 / 38	4 / 34	18
ℓ_2-norm ODM	5 / 38	6 / 34	7,129
Sparse ODM	2 / 38	3 / 34	16

V. CONCLUSION

The standard ODM exploits the ℓ_2-norm penalty, which is not suitable when redundant noisy features exist or a computation cheape model is required. In this paper, we propose the Sparse ODM, which can achieve good generalization performance with moderate model size. In the future, we will consider introducing group sparsity inducing penalties.

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